

2026 SOLIDS MECHANICAL

Progress Report — Attachment Team

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1. Introduction

This document is the Solids team's 2026 internal progress ledger for the attachment sub-team. It covers the existing motor casing design, structural calculations, open-motor simulation results, key challenges encountered this season, and the evaluation of alternative casing materials. Update (17 June 2026): Sections 9–12 have been added, covering formalised BATES grain geometry design rules and the structural and performance evaluation of a third candidate casing material, Stainless Steel G316, against the Aluminium and Carbon Steel designs above.

2. Existing Motor Casing Design

2.1 Geometry & Material

The baseline motor casing is fabricated from Aluminium 6063-T5 alloy. The key dimensional and material parameters are listed below.

Material: Aluminium 6063-T5 Alloy
Inner Diameter: 94 mm
Outer Diameter: 100 mm (wall thickness = 3 mm)
Length: 825 mm
Young's Modulus (E): 70,000 MPa
Poisson's Ratio (μ): 0.33
Yield Strength (σ_{yp}): 110 MPa
Ultimate Tensile Strength (σ_{uts}): 150 MPa

2.2 Design Loads

The following design inputs were used across all structural calculations:

- Maximum Expected Operating Pressure (MEOP, P_e): 4.26 MPa
- Safety Factor (Sf): 1.5

2.3 Structural Calculations — Aluminium Casing

Minimum Wall Thickness

Formula: $t = (P_e \times d \times S_f) / (2 \times \sigma_{yp})$
Result: $t = (4.26 \times 94 \times 1.5) / (2 \times 110) = 2.730 \text{ mm} \rightarrow$ **Selected thickness: 3 mm (manufacturing tolerance)**

Expected Change in Length

Formula: $\Delta L = (P_e \times d \times L \times (1 - 2\mu)) / (4 \times t \times E)$
Result: $\Delta L = (4.26 \times 94 \times 725 \times (1 - 2 \times 0.33)) / (4 \times 3 \times 70,000) \rightarrow \approx 0.1379 \text{ mm}$

Expected Change in Diameter

Formula: $\Delta d = (P_e \times d^2 \times (1 - \mu)) / (4 \times t \times E)$
Result: $\Delta d = (4.26 \times 94^2 \times (1 - 0.33)) / (4 \times 3 \times 70,000) \rightarrow \approx 0.0352 \text{ mm}$

Burst Pressure

Formula: $P_B = (2 \times t \times \sigma_{uts}) / d$
Result: $P_B = (2 \times 3 \times 150) / 94 \rightarrow \approx 9.57 \text{ MPa}$ **(safety margin vs MEOP: ~2.25x)**

3. Open Motor Simulation — Baseline Aluminium Design

An open-motor simulation of the propellant grain stack generated the following performance estimates. The configuration consists of five grains, each 145 mm long and 86 mm in diameter for the aluminium design and 96.26 mm in diameter for the carbon steel design, with a stepped-core diameter sequence (bulk-head to aft) of {38, 35, 35, 34, 34} mm.

Metric	Aluminium Design	Carbon Steel Design
Motor Designation	M2458 (58%)	M3458 (38%)
Impulse	8,114.60 Ns	10,855 Ns
Average Pressure	3.98 MPa	4.74 MPa
Peak Pressure	4.42 MPa	5.47 MPa
Delivered ISP	134.92 s	137.30 s

Metric	Aluminium Design	Carbon Steel Design
Burn Time	3.27 s	3.60 s
Propellant Mass	6.13 kg	8.06 kg
Propellant Length	725 mm	725 mm
Thrust	2,458 N	2,990 N
Port/Throat Ratio	2.73	2.73
Volume Loading	83.22%	86.70%

4. Challenges

Two primary challenges have been identified this season:

- Material sourcing (primary challenge): Aluminium 6063-T5 stock with the required inner or outer diameter is not readily available from local Kenyan suppliers. This was the principal factor behind previous failed static tests.
- Grain porosity: Prior motors suffered catastrophic failures attributable in part to air cavities within the propellant grain, leading to unpredictable burn behaviour.

5. Team Goals & Responsibilities — 2026 Season

- Identify and qualify a locally available casing material and adapt the design accordingly.
- Improve grain casting procedures to minimise internal porosity and air cavities.
- Conduct a successful static motor test.
- Produce flight-ready motors for the actual launch campaign.

6. Motor Casing Material Selection

Because neither Aluminium 6063-T5 nor 6061-T6 is available locally in dimensions close to those required, two alternative materials were assessed. Machining constraints restrict pipe modifications to turning (OD reduction) or boring (ID enlargement) — not both simultaneously.

6.1 Option A — Mild Steel (Existing Inventory)

Available Stock (Nakuja inventory):

- Inner Diameter: 89.03 mm
- Outer Diameter: 115 mm
- Cut Length: 750 mm

Machining plan: Hold ID constant; turn OD down to 94 mm for a wall thickness of ~2.5 mm

Liner thickness: 3 mm → effective grain diameter: 83 mm

Grain height: 130 mm (5 grains)

Simulation Results (Mild Steel)

- Impulse: 6,525 Ns (vs. 8,114.60 Ns baseline — 20% reduction)
- Thrust: 2,014 N (vs. 2,458 N baseline — 18% reduction)
- Port/Throat Ratio: 2.73

Assessment

- **Advantages:** Already in inventory; can be turned to fit.
- **Disadvantages:** Heavier than aluminium; pre-cut length limits usable grain volume despite available casing height; lower impulse than baseline.
- **Verdict:** Not recommended. The combined effect of reduced thrust (2,014 N vs 2,458 N) and higher casing mass would significantly compromise the target apogee.

6.2 Option B — Grade B API 5L Schedule 40 Carbon Steel

Off-the-shelf dimensions:

- Outer Diameter: 114.3 mm
- Inner Diameter: 102.26 mm (off-the-shelf)
- Stock Length: 5,800 mm

Material properties:

- Yield Strength (σ_{yp}): 241 MPa (SMYS)
- UTS (σ_{uts}): 415 MPa (minimum)

- Young's Modulus (E): 200,000 MPa
- Poisson's Ratio (μ): 0.30

Structural Calculations — Carbon Steel Casing

Min. Thickness: $t = (4.26 \times 102.26 \times 1.5) / (2 \times 241) \rightarrow \approx 1.34 \text{ mm}$
→ **Selected: 2 mm**

ΔL : $\Delta L = (4.26 \times 102.26 \times 825 \times (1 - 2 \times 0.3)) / (4 \times 2 \times 200,000)$
→ $\approx 0.090 \text{ mm}$

Δd : $\Delta d = (4.26 \times 102.26^2 \times (1 - 0.3)) / (4 \times 2 \times 200,000) \rightarrow \approx 0.0195 \text{ mm}$

Burst Pressure: $PB = (2 \times 2 \times 415) / 102.26 \rightarrow \approx 16.23 \text{ MPa}$
(safety margin vs MEOP: $\sim 3.81\times$)

Simulation Results (Carbon Steel, grain dia. = 96.26 mm, h = 145 mm)

- Impulse: 10,855 Ns (vs. 8,114.60 Ns baseline — +34%)
- Thrust: 2,990 N (vs. 2,458 N baseline — +22%)
- Port/Throat Ratio: 2.73
- **Note:** Peak mass flux exceeds the 2.0 lb/s/in² threshold for non-erosive burning. Nozzle redesign is required.

Assessment

- **Advantages:** Locally available in Kenya; high burst pressure (16.23 MPa); larger bore increases propellant volume and compensates for added casing mass.
- **Disadvantages:** Heavier than aluminium; peak mass flux exceeds non-erosive limit — nozzle must be redesigned to accomodate the high flow rates of between 2.0 and 3.0.

7. Material Comparison Summary

The table below compares the two candidate materials against the aluminium baseline across all structural and performance metrics.

Parameter	Aluminium 6063-T5	API 5L Gr B Carbon Steel	Notes
Inner Diameter	94 mm	102.26 mm	Steel has larger bore
Wall Thickness	3 mm	2 mm	Steel needs less material
Min. Required t (hoop)	2.73 mm	1.34 mm	Steel $\sim 2\times$ stronger
ΔL (length change)	0.157 mm	0.090 mm	Steel $\sim 1.8\times$ less deformation

Parameter	Aluminium 6063-T5	API 5L Gr B Carbon Steel	Notes
Δd (diameter change)	0.040 mm	0.0195 mm	Steel ~2.1× less change
Burst Pressure	9.57 MPa	16.23 MPa	Steel ~70% higher
Safety Margin vs MEOP	~2.25×	~3.81×	Steel significantly safer
Weight	Lighter	Heavier	Higher density + thickness
Simulated Impulse	8,114.60 Ns	10,855 Ns	Steel ~34% more impulse
Simulated Thrust	2,458 N	2,990 N	Steel ~22% more thrust

8. Recommendation & Next Steps 16 June 2026

Based on the structural analysis and simulation results, the team recommends proceeding with Grade B API 5L Schedule 40 carbon steel as the motor casing material.

RECOMMENDED: Grade B — API 5L Schedule 40 Carbon Steel

Action items:

1. Procure API 5L Gr B Schedule 40 pipe stock (OD 114.3 mm, ID 102.26 mm).

2. Turn casing wall to a final thickness of 1.5 – 2.0 mm.

3. Maintain inner casing diameter at 102.26 mm.

4. Determine final grain diameter after liner thickness is confirmed.

5. Redesign the nozzle to bring peak mass flux below 2.0 lb/s/in².

This eliminates the risk of erosive burning, over-pressurisation, and CATO.

9. BATES Grain Geometry & Erosive Burning Design Limits

Before evaluating Stainless Steel G316 as a casing material, the team formalised three geometric design rules that govern safe BATES grain combustion. These rules were applied retroactively to flag a margin issue in the existing aluminium-based design and were used going forward to size the G316 grain stack.

9.1 Grain Length-to-Diameter Ratio (Neutrality)

Each individual BATES grain segment burns from the inner core outward (progressive, as internal surface area increases) and from its two flat ends inward (regressive, as the segment shortens). These two effects cancel almost exactly when the segment length is approximately 1.5 times its outer diameter, producing a near-neutral thrust profile.

Lg / Do ≈ 1.5

Lg = length of a single grain segment | Do = outer diameter of the propellant grain

References

([U.S. Air Force Rocket Propulsion Laboratory and United Technology Center](#) #Pg 4-5)
(Davenas #Chapter 2) [Access this book here](#)

9.2 Port-to-Throat Area Ratio (Erosive Burning Safety)

When multiple BATES segments are stacked into a single motor, the ratio of core cross-sectional area (AP) to nozzle throat area (AT) sets the absolute ceiling on how much gas can travel down the core before erosive burning becomes dangerous. This is treated as the single most safety-critical metric for a KNSB motor of this scale.

AP / AT = (core cross-sectional area) / (nozzle throat cross-sectional area)
2.0 to 3.0 – acceptable range, scaled by motor length

- **Short motors (2–3 grains):** an AP/AT of 2.0 is acceptable.
- **Long motors (4–6 grains):** AP/AT of 2.5–3.0 is required to accommodate the larger accumulated gas volume travelling down the core provide peak pressures are below 4.14 Mpa (Kloeblen)

References

(Sutton and Biblarz) [Access this book here](#)
(Kloeblen) [Access this Applied Engineering Citation](#)

9.3 Port Length-to-Diameter Ratio (Overall Core Length Limit)

If total grain stack length is too long relative to grain diameter, gas generated at the forward end of the motor must travel a long distance down the core before reaching the nozzle, accelerating to velocities that drive severe erosive burning and chamber pressures beyond the casing's structural limit.

- Standard range for 3–4 grain KNSB BATES motors: L/D between 4.5 and 6.0.
- Absolute maximum: L/D = 7.5. Crossing this threshold with standard straight BATES grains makes erosive burning practically unavoidable unless the core is deliberately widened well past the normal standard, which wastes propellant volume. For stepped cores 8.5 is the absolute maximum

Design flag: the previous aluminium design's port length-to-grain-diameter ratio was 8.4 — close the 8.5 maximum for stepped core designs.

References

Nakka, R. (2020). *KNPSB propellant chemistry and performance characteristics*.
Richard Nakka's Experimental Rocketry Site. nakka-rocketry.net

10. Stainless Steel G316 — Candidate Casing Material

A third candidate material, Stainless Steel Round Tube Grade 316, was identified for sourcing from Ken's Metal Industries. Its properties were compared against the Grade B API 5L Schedule 40 carbon steel evaluated on 16 June 2026 (Section 6.2) and against the original Aluminium 6063-T5 baseline (Section 2).

10.1 Geometry & Material Properties

Material: Stainless Steel G316 (Austenitic, molybdenum-bearing for superior corrosion resistance)

Outer Diameter: 101.6 mm (off-the-shelf)

Inner Diameter: 97.6 mm (off-the-shelf) → casing wall thickness = 2.0 mm

Stock Length: 5,800 mm

Liner thickness (insulation, inside casing): 3 mm → effective grain outer diameter = 91.6 mm

Yield Strength (σ_{yp}): 205 MPa (SMYS)

Ultimate Tensile Strength (σ_{uts}): 515 MPa (minimum)

Young's Modulus (E): 200,000 MPa

Poisson's Ratio (μ): 0.30

10.2 Grain Stack Sizing Against the Section 9 Design Rules

With a grain outer diameter of 91.6 mm (after liner), the Section 9 ratios were used to size the grain stack:

- **Max port length ($L/D \leq 7.5$):** $91.6 \times 7.5 \approx 687$ mm
- **Max single-grain length ($L_g/D_o \approx 1.5$):** $91.6 \times 1.5 \approx 137$ mm

Selected configuration: 5 grains at 130 mm each.

- **Grain geometry ratio:** $130 / 91.6 \approx 1.4$ — slightly below the 1.5 neutral point, giving a mildly regressive burn (a small thrust tail-off near the end of burn). This is treated as desirable for a long motor since it naturally relieves chamber pressure as the casing heats up.
- **Port length:** $130 \times 5 = 650$ mm
- **Port L/D ratio:** $650 / 91.6 \approx 7.1$, rounded to 7.0 for reporting — comfortably under the 8.5 ceiling and a meaningful improvement over the aluminium design's 8.4.

Why not push closer to the 8.5 limit: at this 91.6 mm grain diameter, KNSB propellant shrinks slightly as it cools after casting, which can leave small structural gaps or micro-cracks. Designing right at the geometric maximum leaves no margin — a minor casting imperfection could trigger an ignition pressure spike. The 7.0–7.1 ratio leaves room for that variability.

11. Structural Calculations & Simulation — G316 Casing

11.1 Open Motor Simulation

Holding the same stepped core diameter configuration used in the aluminium design (nozzle-end to bulkhead: 38, 35, 35, 34, 34 mm), an open-motor simulation of the five-grain, 91.6 mm diameter G316 stack produced the following:

Metric	Aluminium Baseline	Stainless Steel G316
Impulse	8,114.60 Ns	8,347.56 Ns
Thrust	2,458 N	2,226 N
Port/Throat Ratio	2.73	2.73
MEOP (Pe)	4.26 MPa	4.06 MPa

Impulse is marginally higher than the aluminium baseline, but thrust is lower than both the aluminium baseline (2,458 N) and the carbon steel design (2,990 N, Section 3). This trade-off is a direct consequence of designing strictly within the Section 9 erosive-burning limits rather than pushing grain geometry as far as the carbon steel design did.

11.2 Structural Calculations

Min. Thickness: $t = (Pe \times d \times Sf) / (2 \times \sigma_{yp}) = (4.06 \times 97.6 \times 1.5) / (2 \times 205) \approx 1.45 \text{ mm} \rightarrow \text{Selected: } 2.0 \text{ mm} \text{ (~1.38\times minimum required)}$

ΔL (length change): $\Delta L = (Pe \times d \times L \times (1 - 2\mu)) / (4 \times t \times E) = (4.06 \times 97.6 \times 750 \times (1 - 0.6)) / (4 \times 2 \times 200,000) \approx 0.074 \text{ mm}$

Δd (diameter change): $\Delta d = (Pe \times d^2 \times (1 - \mu)) / (4 \times t \times E) = (4.06 \times 97.6^2 \times 0.7) / (4 \times 2 \times 200,000) \approx 0.017 \text{ mm}$

Burst Pressure: $PB = (2 \times t \times \sigma_{uts}) / d = (2 \times 2.0 \times 515) / 97.6 \approx 21.11 \text{ MPa} \text{ (safety margin vs MEOP: ~5.2\times)}$

11.3 Thrust-to-Weight Ratio

Design	Thrust (N)	Rocket Weight (N)	Thrust : Weight
Aluminium 6063-T5	2,458	247.7	9.93 : 1
Stainless Steel G316	2,226	286.7	7.76 : 1

A thrust-to-weight ratio between 5:1 and 10:1 is considered acceptable for this airframe. G316's 7.76:1 clears that bar but sits below the aluminium baseline, driven by the casing's higher density.

12. Three-Way Material Comparison & Recommendation

The table below extends the Section 7 comparison to include Stainless Steel G316 alongside the Aluminium 6063-T5 baseline and the API 5L Gr B Carbon Steel candidate.

Parameter	Aluminium 6063-T5	Mild steel (In stores)	API 5L Gr B Mild Steel	Stainless Steel G316
Available length	-	725 mm	-	-
Outer Diameter	100 mm	115 mm	114.3 mm	101.6 mm
Inner Diameter	94 mm	89.03 mm	102.26 mm	97.6 mm
Liner thickness	3 mm	3 mm	3 mm	3 mm
Grain Diameter	88 mm	83 mm	96.26 mm	91.6 mm
Peak Pressure	4.26 MPa	3.58 MPa	6.63 MPa	4.63 MPa
Grain Length	145 mm	130 mm	145 mm	140 mm
No. of Grains	5	5	5	5
Grain Steps (Bulk head to nozzle)	(34,34,35,35,38)	(34,34,35,35,38)	(34,34,35,35,38)	(34,34,35,35,38)
Yield Strength	110 MPa	250 MPa	245 MPa	205 MPa
UTS	150 MPa	400 MPa	415 MPa	515 MPa
Youngs Modulus	70 GPa	200 GPa	200 GPa	200 GPa
Min. Required t [sf = 1.5]	2.73 mm	1.2 mm	2.1 mm	1.7 mm
Min. Required t [sf = 2.0]	3.6 mm	1.6 mm	2.8 mm	2.2 mm
Selected Thickness	3.0 mm (fixed)	2.5 mm	2.5 mm	2.0 mm (fixed)
Final Outer Diameter	100 mm	91.53 mm	104.76 mm*	101.6 mm
Burst Pressure	9.57 MPa	22.46 MPa	20.29 MPa	21.11 MPa
Safety Margin vs MEOP	~2.25×	~6.27×	~3.10×	~4.56×
Simulated Impulse	8,114.60 Ns	6,525.66 Ns	10,768.32 Ns	9135.88 Ns
Simulated Thrust	2,458 N	2,014 N	2,966 N	2,537 N
Density	~ 2800 kg/m ³	~ 8000 kg/m ³	~ 8000 kg/m ³	~ 8000 kg/m³
Casing Length	825 mm	750 mm	825 mm	800 mm
Casing Volume ($\times 10^{-6}$)	754.22 m ³	265.90 m ³	335.35 m ³	500.64 m³
Casing Mass	2.11 kg	2.13 kg	2.68 kg	4.01 kg
Propellant Mass	6.48 kg	5.05 kg	8.00 kg	6.86 kg
Motor Mass (Bulk head mass ~594g) (Nozzle mass ~3500g)	~12.68 kg	~11.27 kg	~14.77 kg	~14.96 kg

Parameter	Aluminium 6063-T5	Mild steel (In stores)	API 5L Gr B Mild Steel	Stainless Steel G316
Simulated Apogee				~2922 m
Thrust : Weight	9.93 : 1		—	7.76 : 1
Local Availability (Kenya)	Limited	In Stores	Available	Available

12.1 Assessment

- **Advantages:** Highest burst pressure and safety margin of the three candidates (~5.2× MEOP); off-the-shelf dimensions require no machining; no nozzle or core redesign needed; superior corrosion resistance versus carbon steel; the only candidate designed within the full Section 9 erosive-burning limits (Port L/D = 7.0 vs the carbon steel design's unresolved nozzle redesign requirement).
- **Disadvantages:** Lower thrust than both other candidates, driven by conservative grain geometry rather than material limitations; heavier than aluminium, reducing thrust-to-weight ratio and projected apogee; cost not yet confirmed with the supplier.

12.2 Conclusion

Stainless Steel G316 tube, 4" (101.6 mm) OD with a 2 mm wall thickness, is usable as a casing material with no additional machining, no nozzle redesign, and no body tube redesign. The resulting motor still qualifies as M-class despite the reduction in thrust, and simulated burn time increased from 3.27 s (aluminium) to 3.72 s. This is a direct result of designing the grain stack for smoother, non-erosive combustion under the Section 9 geometry rules, rather than maximising thrust. The aluminium design's Port L/D ratio of 8.4 close the 8.5 safety threshold; the G316 design's ratio of 7.0 corrects this and increases confidence in the motor's probability of a successful burn. The main open concern is the impact of the added casing weight on projected apogee.

RECOMMENDATION — Next Steps 17 June 2026

1. Contact Ken's Metal Industries for a firm quote on Stainless Steel G316 tube stock.
2. Obtain a comparable quote for API 5L Gr B carbon steel, inclusive of machining to size, for a direct cost comparison.
3. Model the impact of the increased G316 casing weight on projected apogee.
4. Confirm final grain outer diameter (91.6 mm) once liner thickness is verified against the as-received tube ID.
5. Carry forward the Section 9 BATES geometry rules as a standing design check for all future grain stack revisions if a successful static test is achieved

Works Cited

1. *Richard Nakka's Experimental Rocketry Site*, <http://nakka-rocketry.net>. Accessed 17 June 2026.
2. Davenas, Alain, editor. *Solid Rocket Propulsion Technology*. Elsevier Science & Technology Books, 1993. Accessed 17 June 2026.

3. Kloeblen, William R. “Erosive Burning Design Criteria for High Power and Experimental Solid Rocket Motors.” *Richard Nakka Rocketry Research Portal*, https://www.nakka-rocketry.net/articles/Erosive_Burning_Design_Criteria.pdf. Accessed 17 June 2026.
4. Sutton, George P., and Oscar Biblarz. *Rocket Propulsion Elements*. Wiley, 2016. Accessed 17 June 2026.
5. U.S. Air Force Rocket Propulsion Laboratory, and United Technology Center. “BALLISTIC MISSILE PROPELLANT EVALUATION TEST MOTOR SYSTEM (SUPER BATES).” Technical Report AFRPL-TR-74-63 details the Super BATES (Ballistic Test and Evaluation System), a standardized, case-bonded solid propellant system developed for high-precision motor performance measurements. This 1974 Air Force Rocket Propulsion Laboratory. *TECHNICAL REPORT - AFRPL-TR-74-63*, NATIONAL TECHNICAL INFORMATION SERVICE - U. S. DEPARTMENT OF COMMERCE, 25 November 1974, <https://apps.dtic.mil/sti/tr/pdf/ADA004218.pdf>. Accessed 17 June 2026.

— End of Report —